

CS 217: Practice questions on clustering

1. Consider the following eight points with $d = 3$ binary attributes. Assume that we measure distance between two points $\mathbf{x}^i$ and $\mathbf{x}^j$ as $d(\mathbf{x}^i, \mathbf{x}^j) = \sum_{k=1}^d |x_k^i - x_k^j|$.

	x_1	x_2	x_3
$\mathbf{x}^0$	0	0	0
$\mathbf{x}^1$	0	0	1
$\mathbf{x}^2$	0	1	0
$\mathbf{x}^3$	0	1	1
$\mathbf{x}^4$	1	0	0
$\mathbf{x}^5$	1	0	1
$\mathbf{x}^6$	1	1	0
$\mathbf{x}^7$	1	1	1

- (a) Show the clustering at the end of an iteration of K-means where the number of clusters $K = 2$ and the starting centroids are at $m_1 = [1 \ 0 \ 0]$ and $m_2 = [0 \ 1 \ 1]$

- Members of cluster 1:

x_0, x_4, x_5, x_6

- Members of cluster 2:

x_1, x_2, x_3, x_7

..1

- (b) Now we will apply the EM algorithm on this data. Assume the probability of an instance in each cluster follows an independent Bernoulli distribution that is, $\Pr(\mathbf{x}|k) = B(p_{k1}, x_1)B(p_{k2}, x_2)B(p_{k3}, x_3)$ where $B(p, x) = p^x(1-p)^{1-x}$ is the probability of x under a Bernoulli distribution with parameter p . For example, $\Pr(\mathbf{x}^3|k) = (1 - p_{k1})p_{k2}p_{k3}$.

- i. Say at iteration t $[p_{11}^t \ p_{12}^t \ p_{13}^t]$ is $[0.5 \ 0.5 \ 0.1]$ and $[p_{21}^t \ p_{22}^t \ p_{23}^t]$ is $[0.5 \ 0.5 \ 1]$ and $\pi_1^t = \pi_2^t = 1/2$. Write the z_{ik}^t values for each instance in this table. [Note: $z_{ik} = \Pr(k|\mathbf{x}^i, \theta^t)$]

	z_{i1}^t	z_{i2}^t
$\mathbf{x}^0$	1	0
$\mathbf{x}^1$	$\frac{1}{1.1}$	$\frac{0.1}{1.1}$
$\mathbf{x}^2$	1	0
$\mathbf{x}^3$	$\frac{1}{1.1}$	$\frac{0.1}{1.1}$
$\mathbf{x}^4$	1	0
$\mathbf{x}^5$	$\frac{1}{1.1}$	$\frac{0.1}{1.1}$
$\mathbf{x}^6$	1	0
$\mathbf{x}^7$	$\frac{1}{1.1}$	$\frac{0.1}{1.1}$

..3

- ii. What will be the value of p_{13}^{t+1} and p_{23}^{t+1} at the end of the M step of this iteration.

$$p_{23}^{t+1} = 1$$

$$p_{13}^{t+1} = \frac{4 * \frac{0.1}{1.1}}{4 * \frac{0.1}{1.1} + 4 * 1} = \frac{0.1}{1.2} < 0.1$$

..2

- iii. What will be the final converged values of the parameters at the end of all EM iterations. [Justify briefly]

With each iteration, p_{13}^{t+1} reduces and finally converges to 0.

$$p_{23}^{t+1} = 1 \quad \dots 2$$

- iv. Do you see a nice description of the two kinds of clusters that the algorithm produces [Hint: The instances denote binary representation of numbers]

All odd numbers go to one cluster, and even numbers go to another cluster. $\dots 1$

- v. What is another set of parameters where the EM algorithm achieves local optimality. [Justify briefly].

$$p_1^0 = [0.5 \ 0.5 \ 0.5]$$

$$p_2^0 = [0.5 \ 0.5 \ 0.5] \quad \dots 2$$

2. Consider the following dataset with two attributes x_1 and x_2 with value x_2 of instance x^4 missing. We will use EM to fit the most likely value of the missing attribute assuming the data follows a 2-D Gaussian distribution:

$$f(x_1, x_2) = \frac{1}{2\pi\sigma_1\sigma_2\sqrt{1-\rho^2}} \exp -\frac{1}{2(1-\rho^2)} \left(\frac{(x_1-\mu_1)^2}{\sigma_1^2} + \frac{(x_2-\mu_2)^2}{\sigma_2^2} - \frac{\rho(x_2-\mu_2)(x_1-\mu_1)}{\sigma_1\sigma_2} \right)$$

	x_1	x_2
x^1	2	2
x^2	0	0
x^3	0	0.5
x^4	-2	?

- (a) Initially assume that x_1 and x_2 are independent of each other. What value for the missing information would you assign in such a case? Why? $\dots 2$

When the attributes are independent of each other, knowing the value of x_1 gives us no information about x_2 . The best we can do is assign the MLE from the parameters estimated using the first three points. That is just the average $(2+0+0.5)/3$.

- (b) Now, for the case where x_1 and x_2 are correlated. Which parameters from $\mu_1, \mu_2, \sigma_1, \sigma_2, \rho$ would change from one iteration of EM to the next? $\dots 1 \quad \mu_2, \sigma_2 \text{ and } \rho$

- (c) Calculate the maximum likelihood estimates of the parameters that remain fixed across EM iterations. $\dots 1 \quad \text{Easy to calculate MLE of } \mu_1 \text{ and } \sigma_1$

- (d) Show that the conditional probability $\Pr(x_2|x_1 = c)$ is a Gaussian distribution whose mean is a function of c . $\dots 3$

- (e) At step $t+1$ of EM, write the expression for the E step in terms of the value of the fixed parameters above, and the value of the changing parameters at step t (for example, use σ_2^t to denote the value of σ_2 at step t). $\dots 2$

- (f) At step $t+1$, write the equations for the M step of EM. $\dots 3$

3. Consider the following estimation problem. Your friend has two coins with probability of coming up heads for coin 1 = p_1 and coin 2 = p_2 . He performs this experiment four times: choose coin 1 with probability λ or coin 2 with probability $1-\lambda$ and toss the chosen coin three times. Let the observed outcome of coin tosses be $Y = \{\{HHH\}, \{TTT\}, \{HHH\}, \{TTT\}\}$. Let z_i for $i = 1, 2, 3, 4$ denote the coin chosen in each experiment.

- (a) First, consider the case where the value of z_i is known and is observed to be $(z_1, z_2, z_3, z_4) = (1, 2, 1, 1)$. Write down the maximum likelihood estimation of the parameters.

- i. $\lambda =$..1
 - ii. $p_1 =$..1
 - iii. $p_2 =$..1
- (b) Now consider the case where z_i is not known. We will use EM to estimate the parameters (λ, p_1, p_2) . Let $(\lambda^t, p_1^t, p_2^t)$ denote the value of the parameter at the beginning of the t -th iteration. For the **E** step of EM, express in full expanded form the expected value of z_i in terms of these and the absolute counts h_i of heads in the i -th experiment in Y . ..3
- (c) For the **M** step, express the updated parameter values, in terms of expected values above and the absolute counts of Hs and Ts in each element of Y .
 - i. $\lambda =$..1
 - ii. $p_1 =$..2
 - iii. $p_2 =$..1
- (d) Identify an assignment of $(\lambda^*, p_1^*, p_2^*)$ at which EM cannot make progress but which is not globally optimal. You need to provide the value of data likelihood at this assignment and another assignment for which likelihood is higher and show that the **E** and **M** steps cause no change at $(\lambda^*, p_1^*, p_2^*)$. (Hint: try to guess the optimal parameter values from the Y s) ..4